



Habib University
shaping futures

Computational Intelligence

CS 451 L1

Spring Semester 2026

Course Information

Start Date	End Date	Class Location	Meeting Time
Jan 12 2026	Feb 18 2026	GF-C109	Tues Thurs (05:15 PM-06:30 PM)
Feb 19 2026	Mar 20 2026	Tues Thurs (03:00 PM-04:00 PM)	Tues Thurs (03:00 PM-04:00 PM)
Mar 24 2026	Apr 30 2026	GF-C109	Tues Thurs (05:15 PM-06:30 PM)

"In all things of nature there is something of the marvellous"

Aristotle

"Look deep into nature, and then you will understand everything better"

Albert Einstein

Course Prerequisites: EE/CE 354/361,MATH 310

Hardware/Software Prerequisites (if any).

Content Area: This course serves as a higher-level CS elective or free elective and also fulfils the requirement of elective for CS minor. Enrolment in the course is subject to meeting the prerequisites.

Instructor Information

Instructor: Syed Mujtaba Hassan Naqvi

Office Location: TBD

Email: mujtaba.hassan@sse.habib.edu.pk

Office Hours: TBD

Instructor: Syeda Saleha Raza

Title: Associate Professor

Office Location: C-212

Email: saleha.raza@sse.habib.edu.pk

Office Hours: Mon (12:00 -1:00 pm), Tue/Thu (4:00 - 5:00 pm), Others: By appointment

Course Description

This is a higher-level CS elective in the stream of Artificial Intelligence (AI) that exposes students to more advanced areas in AI and enable them to address complex optimization problems for which no tractable solution exists.

Course Learning Outcomes (CLOs)

By the end of the course, students will be able to:

1. Analyze real-world problems and model them as optimization problems using appropriate computational techniques.
2. Apply Evolutionary Algorithm, Swarm Intelligence, and Artificial Immune System to solve complex optimization problems, including non-convex and NP-hard problems.
3. Explain the difference between uncertainty and fuzziness, and design fuzzy logic models to handle imprecise information in real-world systems
4. Explain basic principles of cellular automata and their use to model natural and social phenomena
5. Develop the ability to critically read, analyze, and summarize research papers, and write technical reports applying proper scientific methodology.

Course Aims

This course is based on the study of different nature-inspired computational methods. During

the course, students will learn different computational techniques that take inspirations from natural/biological systems. Students will gain hands-on experience of applying these techniques to solve complex optimization problems. The specific topics include Evolutionary computation, Swarm Intelligence, Artificial Neural Networks, Cellular Automata, and Fuzzy logic.

Mode of Instruction

The class will be conducted twice a week for 75 minutes in in-person mode. Students can expect to work for at least 5-6 hours per week outside of the scheduled class for this course.

All course material will be made available on Canvas.

Engagement & Participation Rules

There will be multiple engagement opportunities during the course in the form of assignments, presentations, quizzes, discussions and open questions. Students are required to participate in all of these activities. Student Engagement in the course will be measured through their participation and performance in:

- quizzes
- timely submission of assignments
- timely response to various surveys conducted during the course
- participation in class discussions

Required Texts and Materials



Fundamentals of Natural Computing

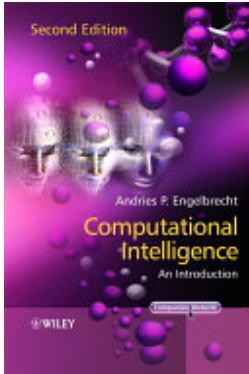
ISBN: 9781584886433

Authors: Leandro Nunes de Castro

Publisher: CRC Press

Publication Date: 2006-06-02

Optional Materials



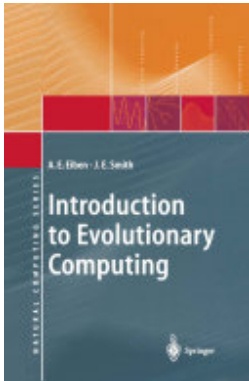
Computational Intelligence

ISBN: 9780470512500

Authors: Andries P. Engelbrecht

Publisher: John Wiley & Sons

Publication Date: 2007-10-22



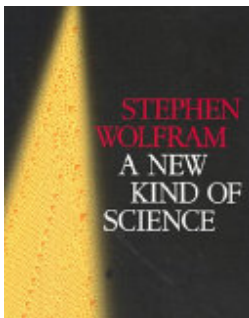
Introduction to Evolutionary Computing

ISBN: 9783662050941

Authors: Agoston E. Eiben, J.E. Smith

Publisher: Springer Science & Business Media

Publication Date: 2013-03-14



A New Kind of Science

Authors: Stephen Wolfram

Publication Date: 2002-01-01

Assessments

Assignments (3)	20%
Quizzes (5)	20%
Presentations	10%

Project + Paper (Proposal - 3% Demo - 12% Paper - 10%)	25%
Participation/Feedback	05%
Final Exam	20%

Grading Scale

Letter Grade	GPA Points	Percentage
A+	4.00	[95-100]
A	4.00	[90-95)
A-	3.67	[85-90)
B+	3.33	[80-85)
B	3.00	[75-80)
B-	2.67	[70-75)
C+	2.33	[67-70)
C	2.00	[63-67)
C-	1.67	[60-63)
F	0.00	[0, 60]

Note: [a, b) is a range of numbers from a to b where a is included in the range and b is not.

Late Submission Policy

Students are required to respect the deadlines given by the instructor and make timely submissions.

If a student is facing some genuine problem that hinders the performance in the assessment, he/she should discuss with the instructor in advance to get some accommodation. No last-minute requests for relaxation will be entertained.

Late submissions without any genuine reason will be subjected to a penalty of 10% grade reduction for every delay of 24 hours.

Week-Wise Schedule (Tentative)

Weekly Schedule

Week	Description	Readings	Assessments and Due Date
Week - 1	Course Overview, Computational Intelligence, Optimization, Evolutionary Computation - Intro	Engelbrecht - Chapter 1	
Week - 2	Evolutionary Algorithms, Survival of the fittest , Exploration vs Exploitation, Problem formulation, Parent Selection & Survivor Selection mechanisms, Using EA to solve Traveling Salesman problem	Engelbrecht - Chapter 8	
Week - 3	Evaluation of EA, Evolutionary Programming and Evolutionary Strategies, Uniform & Gaussian mutations, Genetic Algorithm, Evolutionary Art – Evolving Mona lisa, Evolving bipedal robot walk, Evolving a simulated car	Engelbrecht - Chapter 9,11,12 14 Examples of EA BoxCar2D	Assignment 1 out, Quiz 1 (Optimization)

Week	Description	Readings	Assessments and Due Date
Week - 4	Variants of EA (Evolutionary Programming, Evolutionary Strategies, Genetic Algorithms), Swarms in nature, Key principles of Swarm behaviour	The Gathering of Swarms (Documentary) Swarm Intelligence Engelbrecht (Chapter 16)	Finalize Research paper I to present.
Week - 5	Ant Colony Optimization (ACO)		Assignment 1 due Quiz 2 (EA)
Week - 6	Research Paper Presentations - I	Engelbrecht (Chapter 17)	Project - Problem Due
Week - 7 (Midterm Week)	Research Paper Presentation - I Particle Swarm Optimization (PSO)		
Week - 8 (Midterm Week)	Particle Systems Artificial Neural Networks (ANN) - Recap Evolving Neural Networks,		Quiz 3 (SI) Assignment 2 released,
Week - 9	Self-organizing Maps Project Ideas Discussion		Project - Proposed Solution Due
Week - 10	Fractals Cellular Automata		Assignment 2 due
Week - 11	Cellular Automata - to model natural and social complexity		Project - Literature Review Due Quiz 4 (ANN/ SOM) Assignment 3 Released

Week	Description	Readings	Assessments and Due Date
Week - 12	Artificial Immune System, Fuzzy Logic, Fuzzy variables, operators		Assignment 3 due
Week - 13	Fuzzy Reasoning		Finalize Research Paper II to present.
Week - 14	Research Paper Presentations – II		
Week - 15	Research Paper Presentations – II		Quiz 5 (FL)
Week - 16	Project Demos		Project Submission
	End Term Examinations Days ^S		Project Report Submission

Notes:

* The University reserves the right to correct typographical errors or to adjust the Academic Calendar at any time it deems necessary.

† Subject to the sighting of the moon.

‡ No Class(es).

Attendance Policy

Students are expected to attend all classes. However, there is a buffer of 15% of absences to accommodate any unforeseeable circumstances out of their control. If the absences of student exceed this buffer, they will automatically be dropped from the course by the registrar office.

Final Exam Policy

The final exam will be held in-person in the final exam week.

Academic Integrity

Each student in this course is expected to abide by the Habib University Student Honor Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work.

Scholastic dishonesty shall be considered a serious violation of these rules and regulations and is subject to strict disciplinary action as prescribed by Habib University regulations and policies. Scholastic dishonesty includes, but is not limited to, cheating on exams, plagiarism on assignments, and collusion.

- a. Plagiarism: Plagiarism is the act of taking the work created by another person or entity and presenting it as one's own for the purpose of personal gain or of obtaining academic credit. As per University policy, plagiarism includes the submission of or incorporation of the work of others without acknowledging its provenance or giving due credit according to established academic practices. This includes the submission of material that has been appropriated, // bought, received as a gift, downloaded, or obtained by any other means. Students must not, unless they have been granted permission from all faculty members concerned, submit the same assignment or project for academic credit for different courses.
- b. Cheating: The term cheating shall refer to the use of or obtaining of unauthorized information in order to obtain personal benefit or academic credit.
- c. Collusion: Collusion is the act of providing unauthorized assistance to one or more person or of not taking the appropriate precautions against doing so.

All violations of academic integrity will also be immediately reported to the Student Conduct Office.

You are encouraged to study together and to discuss information and concepts covered in lecture and the sections with other students. You can give "consulting" help to or receive "consulting" help from such students. However, this permissible cooperation should never involve one student having possession of a copy of all or part of work done by someone else, in the form of an e-mail, an e-mail attachment file, a diskette, or a hard copy.

Should copying occur, the student who copied work from another student and the student who gave material to be copied will both be in violation of the Student Code of Conduct.

If you wish to use generative-AI tools to complete any of your assessments, you must first obtain permission from your course instructor. AI generated work will not be accepted in all classes or even all assessments. The instructor's permission is required. If the permission is granted, you should declare its use and properly cite the source of the generated content. Failing to identify AI written or assisted work is academic dishonesty and will be treated as any case of plagiarism by the university.

The principle for academic integrity is that your submissions must be substantially your own work and that any work that is not originally your thought must be identified and credited. If the use of AI tools is prohibited in the course, respect the rules and do not use these tools for assessments. The fundamental purpose of assessment is to learn, synthesize information and explain new connections and interpretations that arise from your secondary research. Be aware that unauthorized use of AI tools for assessments can result in a conduct case being filed. This can have serious consequences for your academic standing and future career opportunities.

During examinations, you must do your own work. Talking or discussion is not permitted during the examinations, nor may you compare papers, copy from others, or collaborate in any way. Any collaborative behavior during the examinations will result in failure of the exam, and may lead to failure of the course and University disciplinary action.

Penalty for violation of this Code can also be extended to include failure of the course and University disciplinary action.

The use of AI tools to complete assessments is strictly prohibited. The instructor reserves the right to conduct a viva for any assessment conducted in the course.

Program Learning Outcomes (For Administrative Review)

Upon graduation, students will have the following abilities:

- PLO 1: Theoretical Computer Science: recall and apply foundational principles of computer science.
- PLO 2: Application Development: build software systems of varying complexity in light of fundamental computer science principles and any other constraints.
- PLO 3: Analysis and Design: perform technical analysis and design using core computing and mathematical knowledge.
- PLO 4: Systems: apply the knowledge of computing systems.
- PLO 6: Problem-Solving: identify and analyze problems and propose effective computing-based solutions.
- PLO 7: Practical Exposure: make effective use of current tools, technologies, and good industry practices.
- PLO 8: Responsible Citizenship: conduct their computing practice in a manner that is ethical and socially responsible, and corresponds to their distinct sense of identity and service to the community.
- PLO 9: Self-Learning: continuously adapts their skills to the changes taking place around them.

- PLO 10: Design Thinking: apply design thinking principles to the design of a solution.
- PLO 11: Multi-disciplinarity: incorporate knowledge and input from multiple disciplines.
- PLO 12: Communication and Teamwork: communicate and function effectively as a member or a leader of a variety of teams.

Program Learning Outcomes (PLOs) mapped to Course Learning Outcomes (CLOs)					
	CLOs of the course are designed to cater following PLOs: PLO 3: Analysis and Design PLO 5: Research and Exploration PLO 6: Problem-Solving PLO 9: Self-Learning				
	Distribution of CLO weightages for each PLO				
	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 3					
PLO 5					
PLO 6					
PLO 9					

Mapping of Assessments to CLOs

Assignments	CLO #01	CLO #02	CLO #03	CLO #04	CLO #05
Assignment 01					
Assignment 02					
Assignment 03					
Quizzes					
Presentations					
Project					
Final Exam					

Recording Policy

Only asynchronous and synchronous online sessions will be conducted and recorded via MS Teams. Link to the recordings will be available to all students on Canvas Learning Management System.

Accommodations for Students with Disabilities

In compliance with the Habib University policy and equal access laws, I am available to discuss appropriate academic accommodations that may be required for student with disabilities. Requests for academic accommodations are to be made during the first two weeks of the semester, except for unusual circumstances, so arrangements can be made.

Students are encouraged to register with the Office of Academic Performance to verify their eligibility for appropriate accommodations.

Inclusivity Statement

We understand that our members represent a rich variety of backgrounds and perspectives. Habib University is committed to providing an atmosphere for learning that respects diversity. While working together to build this community we ask all members to:

- share their unique experiences, values and beliefs
- be open to the views of others
- honor the uniqueness of their colleagues
- appreciate the opportunity that we have to learn from each other in this community
- value each other's opinions and communicate in a respectful manner
- keep confidential discussions that the community has of a personal (or professional) nature
- use this opportunity together to discuss ways in which we can create an inclusive environment in this course and across the Habib community

Office Hours Policy

Every student enrolled in this course must meet individually with the course instructor during course office hours at least once during the semester. The first meeting should happen within the first five weeks of the semester but must occur before midterms. Any student who does not meet with the instructor may face a grade reduction or other penalties at the discretion of the instructor and will have an academic hold placed by the Registrar's Office.